Rediscovering the Higgs Boson

Valen Lebepe

13 August 2025

Abstract

ATLAS Open Data from the LHC were analysed to measure the Z boson mass and search for the Higgs boson in the diphoton decay channel. The Z boson mass was determined as $\boldsymbol{m_Z} = \mathbf{91300} \pm \mathbf{6.83}\ \boldsymbol{MeV/c^2}$, in close agreement with the PDG value. A Higgs signal was observed at $\boldsymbol{m_H} = \mathbf{123\ 934.19} \pm \mathbf{10}\ \boldsymbol{MeV/c^2}$ with a statistical significance of 4.3, demonstrating the effectiveness of event selection and statistical analysis in isolating rare particle decays

Introduction

The ATLAS experiment (A Toroidal LHC Apparatus) at CERN is one of the largest particle physics experiments ever built. At the Large Hadron Collider (LHC), protons are accelerated to nearly the speed of light and collided at centre-of-mass energies up to 13 TeV. Each collision produces a multitude of particles, which are detected, identified, and measured by the ATLAS detector. These collisions occur at a rate of approximately 40 MHz, generating an enormous dataset.

The central challenge of modern experimental particle physics is to isolate the rare but scientifically significant events from this vast background. Once these events are selected, statistical analyses are performed to extract meaningful physical results. In this laboratory investigation, real ATLAS Open Data were analysed to measure the mass of the Z boson and to attempt to rediscover the Higgs boson.

Aim of the Investigation

The aim of this investigation was to measure the mass of the Z boson using muon-pair events from ATLAS Open Data.

Relevant Theory

In high-energy proton–proton collisions at the LHC, a wide range of particles can be produced depending on the energy of these collisions. Among these particles is the Z boson. The Z boson has a very short lifetime and decays almost instantaneously, making direct observation impossible. Instead, its presence is inferred from its decay products.

One of the cleanest and most experimentally accessible channels is the decay of the Z boson into a pair of oppositely charged muons ($Z \rightarrow \mu^+\mu^-$). Muons are relatively easy to identify within the ATLAS detector because they leave distinctive tracks in the outer muon spectrometer and are less affected by strong interactions. Nonetheless, muons can also arise from other processes, such as heavy hadron decays, which must be filtered out. For this reason, event selection criteria needs to be imposed to a dataset of collision events: the muons must be oppositely charged, isolated from nearby hadronic activity, have a large transverse momentum, and lie within a low pseudorapidity region of the detector (small $|\eta|$).

Once suitable events are identified, the invariant mass of the muon pair can be reconstructed from the kinematic information of the muons that the ATLAS detector detects. Energy and momentum conservation dictate that the invariant mass of the system is given by

$$m^2 = E^2 - |\vec{p}|^2$$

where **E** is the total energy and $\vec{\boldsymbol{p}}$ is the total momentum of the muon pair. Ideally, this reconstructed invariant mass corresponds to the Z boson mass, $\boldsymbol{m_Z}$ .However, due to the short lifetime of the Z boson and the uncertainty principle, the measured invariant mass values are not sharply defined. Instead, they follow a Breit–Wigner (or Cauchy) probability distribution,

$$\sigma(E_{\mu^+\mu^-}) \propto \frac{1}{\left(E_{\mu^+\mu^-} - m_Z\right)^2 + \left(\frac{\Gamma_Z}{2}\right)^2}$$

By fitting this distribution to the invariant mass spectrum, one can extract an experimental estimate of $\boldsymbol{m_Z}$. The peak of this distribution corresponds to the best estimate of $\boldsymbol{m_Z}$. In practice, this value is determined by fitting the Cauchy model to the measured $\mathbf{m_{\mu+\mu-}}$ distribution and finding the value of $\boldsymbol{m_Z}$ that minimises the chi-squared difference between the model and the data. This statistical procedure forms the cornerstone of the analysis used in this investigation to determine the Z boson mass.

Methodology

Data and Measurements

The Large Hadron Collider (LHC) collides counter-rotating proton beams at the centre of the ATLAS detector, producing showers of new particles. The detector records millions of such events per second, generating vast datasets. In this experiment, ATLAS Open Data of these events were used, and the analysis was carried out in Python within Jupyter Notebooks, employing libraries such as NumPy, Pandas, Matplotlib, and SciPy.

Measuring the Z Boson Mass

To measure the Z boson mass, events containing two muons were selected from the ATLAS dataset. Selection criteria were applied to reduce background: the muons were required to have opposite charges, transverse momentum larger than a certain value ($p_T > x$), and low pseudorapidity (|η|).After these cuts, a refined dataset was obtained. Using Python within Jupyter Notebooks, the invariant mass of the muon pairs was calculated, and histograms of the distribution were produced. A Cauchy model was then fitted to the data to extract an estimate of $\boldsymbol{m_Z}$.

Results, Analysis and Interpretation

The original dataset contained 7,689,598 entries corresponding to 549,257 unique events. To isolate muons likely originating from Z boson decays, three selection criteria were applied. First, only muons of opposite charge were selected to match the neutral decay signature of the Z boson. Second, a transverse momentum cut of $\mathbf{39,990\,\frac{MeV}{c} < p_T < 560,000\,\frac{MeV}{c}}$ was applied to favor higher-energy muons, thereby increasing the probability that they originated from Z boson decays. Finally, only muons within the pseudorapidity range $\mathbf{0.2 <| \eta |< 2.20}$ were included to ensure that the particles were detected in the most sensitive region of the detector, where kinematic properties can be measured with greater accuracy and background events are effectively filtered out. No isolation criteria were applied. After applying these cuts, the dataset was reduced to 1,732,276 entries, meaning that 77.46% of the original events were excluded.

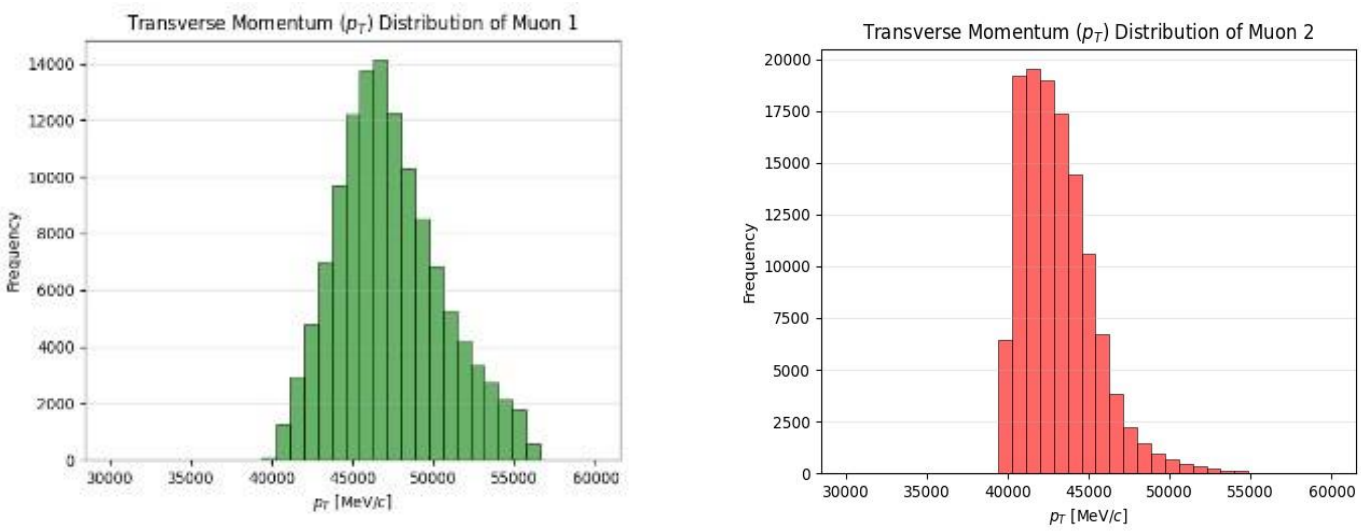

Figure 1:Momentum distribution of muon 1 (left) and muon 2 (right) after applying the selection criteria of opposite charge, transverse momentum $39{,}990 MeV/c < p_T < 560{,}000 MeV/c$, *and pseudorapidity* $0.2 <| \eta |< 2.20$.

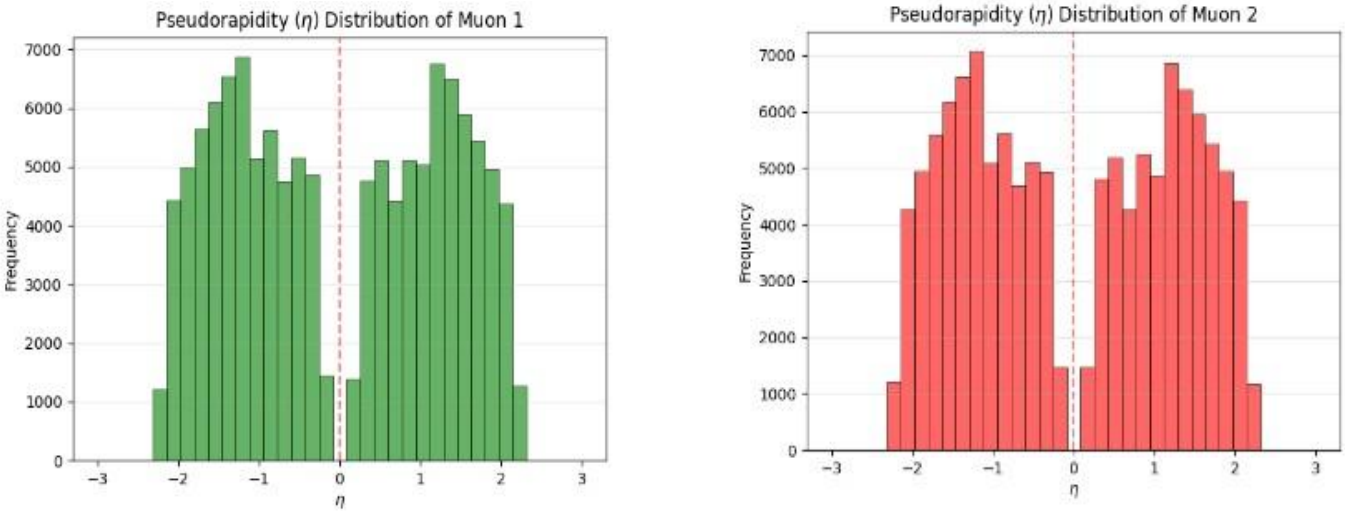

Figure2: Pseudorapidity distributions of muon 1 (left) and muon 2 (right), showing that both distributions are nearly identical due to the back-to-back emission property of Z boson decays.

The momentum and pseudorapidity distributions of the selected muons indicate that the selection effectively isolates the desired events. The pseudorapidity distributions of muon 1 and muon 2 are nearly identical, reflecting the back-to-back emission characteristic of Z boson decays.

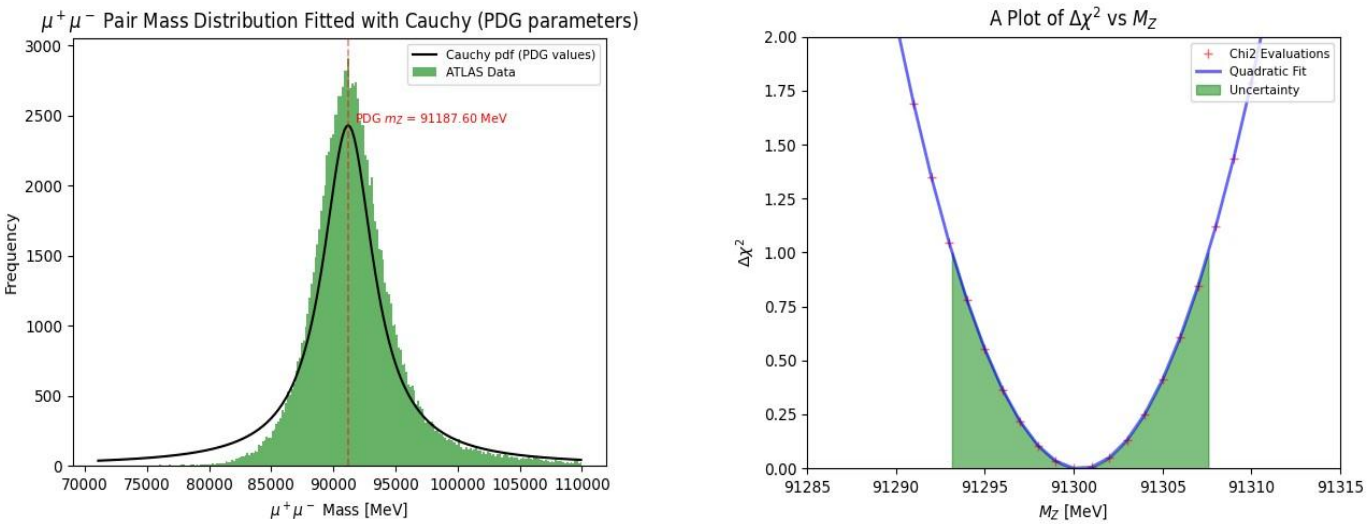

Figure 3: Left: Invariant mass distribution of the muon pair fitted with a Cauchy distribution using the PDG Z boson mass. Right:ΔX^2 versus M_Z showing the best-fit mass that minimizes X^2

Analysis of the invariant mass distribution of the muon pairs shows that while a Cauchy distribution fitted using the PDG Z boson mass ($\boldsymbol{m_{ZPDG}} = \mathbf{91187} \pm \mathbf{4.13}\ \boldsymbol{MeV/c^2}$) does not perfectly align with the data, the peak position is reasonably close. A $\boldsymbol{\chi 2}$ analysis further identifies the best-fit Z boson mass from the dataset to be $\boldsymbol{m_Z} = \mathbf{91300} \pm \mathbf{6.83}\ \boldsymbol{MeV/c^2}$, which differs from the PDG value by 112.4 MeV/c^2. Although this difference is significant, it represents a substantial improvement over the initial selection (default code selection), where the mass deviation was 441.6 MeV/c^2.

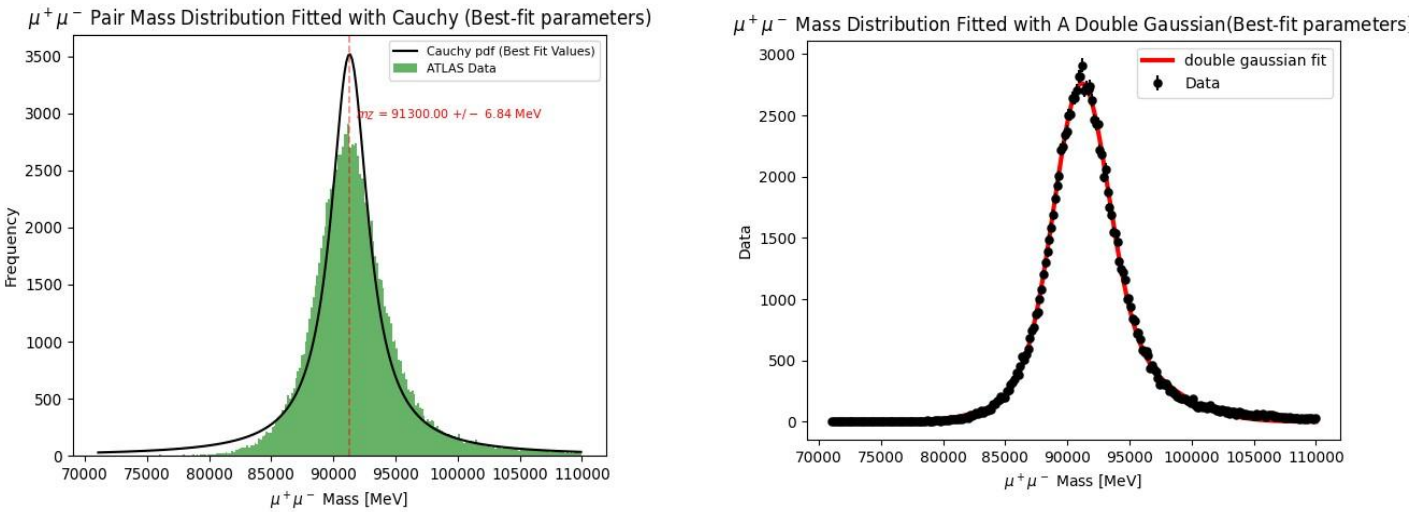

Figure 4: Left: Invariant mass distribution of the muon pair fitted with a Cauchy distribution using the best-fit Z boson mass from the dataset. Right: Invariant mass distribution fitted with a double Gaussian distribution using the best-fit parameters, demonstratrating improved alignment with the data.

Fitting the invariant mass distribution with both a Cauchy distribution using the newly determined Z boson mass and a double Gaussian distribution demonstrates that the double Gaussian provides a slightly better fit, although both distributions capture the peak position accurately.

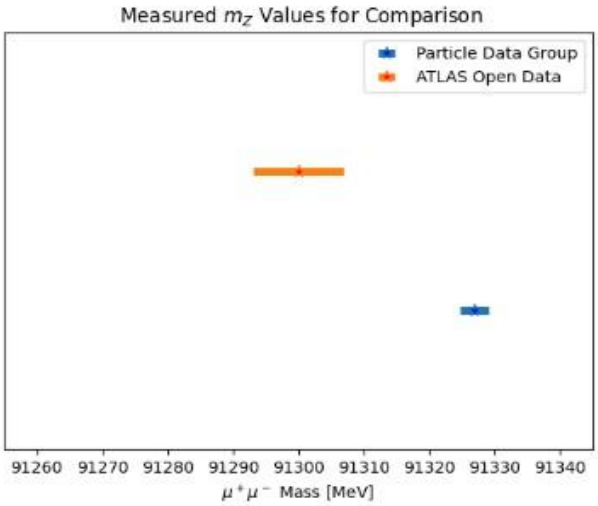

Figure 5: Comparison of the Z boson mass determined from the selected dataset with the PDG value.

Comparing the best-fit Z boson mass obtained from the selection with the PDG value it is observed that the measured mass differs by approximately 112.4 MeV/c^2. Although this difference is larger than the experimental uncertainty, it represents a significant improvement over the initial selection method, where the deviation from the PDG value was 441.6 MeV/c^2.

This demonstrates that the applied selection criteria enhance the accuracy of the Z boson mass determination, though further refinement, such as incorporating particle isolation, could improve the approximation even more.

Conclusion

This investigation successfully measured the Z boson mass from ATLAS Open Data, achieving a best-fit value of $\boldsymbol{m_Z = 91300 \pm 6.83\ MeV/c^2}$, which is closer to the PDG value than the initial selection. Improvement can be made by applying the isolation selection criteria on the muons, which would further reduce background and increase the accuracy of the Z boson mass determination. The Higgs boson was also rediscovered through its diphoton decay channel, with a peak at $\boldsymbol{m_H = 123\,934.19 \pm 10\ MeV/c^2}$, and a statistical significance of **4.3.** While the measured Higgs mass is slightly lower than the PDG value, the analysis demonstrates the feasibility of identifying rare signals in a large dataset using appropriate selection criteria and statistical methods.